| 2 | **Answer: C** |
|---|---|
| | Method 1: Error Propagation<br><br>$a = \frac{b-c}{d^2}$<br><br>Let $K = b - c$, $\therefore K = 21 \pm 4$<br><br>$a = \frac{b-c}{d^2} = \frac{K}{d^2}$<br><br>$\frac{\Delta a}{a} = \frac{\Delta K}{K} + 2\frac{\Delta d}{d}$<br>$= \frac{4}{21} + 2\left(\frac{1}{100}\right)$<br>$= 0.210476 = 0.21\ (2.s.f)$<br><br>Thus, the percentage uncertainty is 21%<br><br>Method 2: Max – Min method<br><br>$a = \frac{b-c}{d^2}$<br><br>Let $K = b - c$, $\therefore K = 21 \pm 4$ |

$$a = \frac{25-4}{100^2}$$

$$a_{\max} = \frac{28-3}{(99)^2}$$

$$a_{\min} = \frac{22-5}{(101)^2}$$

$$\frac{\Delta a}{a} = \frac{\frac{1}{2}(a_{\max} - a_{\min})}{a} = \frac{\frac{28-3}{(99)^2} - \frac{22-5}{(101)^2}}{\frac{25-4}{100^2}}$$

$$= 0.2105373 = 0.21 \text{ (2.s.f)}$$

Thus, the percentage uncertainty is 21%